

## Video 15: *PDE Classification: Elliptic, Parabolic and Hyperbolic Equations*

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# Equation Classification

- Consider the following constant coefficient quadratic equation in 2 variables:

$$Ax^2 + Bxy + Cy^2 = f \quad (1)$$

or

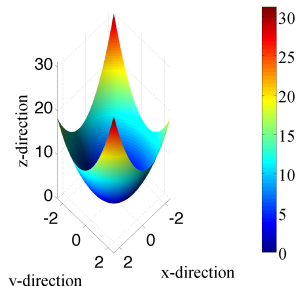
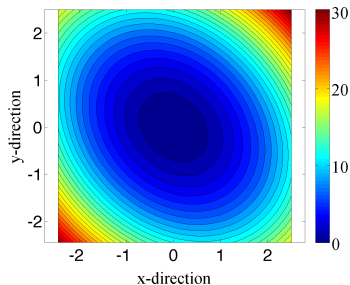
$$A \left( \frac{x}{y} \right)^2 + B \left( \frac{x}{y} \right) + C = \frac{f}{y^2} \quad (2)$$

$$Az^2 + Bz + C = f' \quad (3)$$

- What does this equation represent?
- Let's plot it for several different  $A, B$  and  $C$  values.

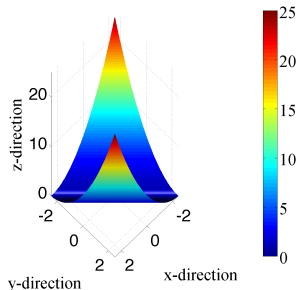
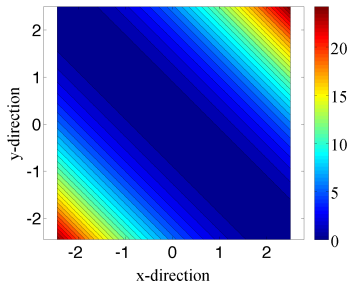
# Equation Classification

- $A = 2, B = 1, C = 2 \rightarrow B^2 - 4AC < 0$
- Paraboloid Surface – Forms an Ellipse



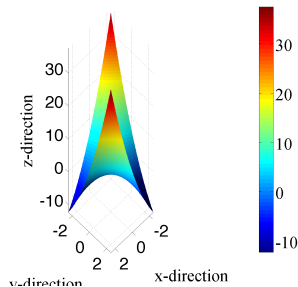
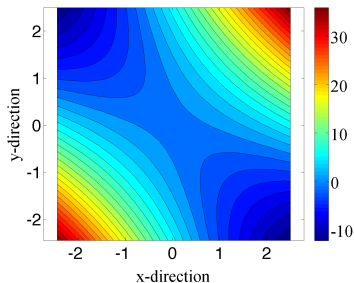
# Equation Classification

- $A = 1, B = 2, C = 1 \rightarrow B^2 - 4AC = 0$
- Pure Parabolic Surface



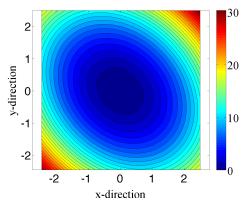
# Equation Classification

- $A = 1, B = 4, C = 1 \rightarrow B^2 - 4AC > 0$
- Hyperbolic Surface

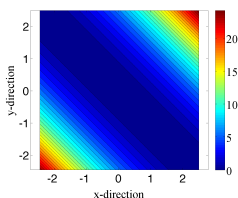


# Summary

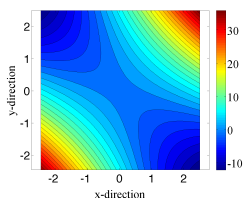
- The value of the discriminant:  $B^2 - 4AC$  defines the equation's behavior
- Contour shapes and the intersection of the shape with the  $x - y$  plane is:
  - $B^2 - 4AC < 0$  : Circles and Ellipse (parabaloid)
  - $B^2 - 4AC = 0$  : Parallel lines (pure parabola)
  - $B^2 - 4AC < 0$  : X-lines (hyperbola)



a) Elliptic



b) Parabolic



c) Hyperbolic

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# PDE Classification - Traditional approach for Linear, CC, 2nd Order PDE classification

- Consider a general, constant coefficient, linear, second order PDE:

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial xy} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu + G = 0 \quad (4)$$

- The classification of a PDE will be governed by the highest-order derivative terms.
- For a second order PDE the classification can be made based on “wave solutions” [Versteeg and Malalasekera 1995].
- The roots of the characteristic equation will depend on the value of the discriminant:

$$\text{discriminant} = (B^2 - 4 \cdot A \cdot C)$$

- The value of the discriminant (as before):
  - $(B^2 - 4 \cdot A \cdot C) > 0 \rightarrow$  Hyperbolic PDE.
  - $(B^2 - 4 \cdot A \cdot C) = 0 \rightarrow$  Parabolic PDE.
  - $(B^2 - 4 \cdot A \cdot C) < 0 \rightarrow$  Elliptic PDE.

# PDE Classification – approach/general concept

- *Aside:* We can show that the characteristic equation:

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial xy} + C \frac{\partial^2 u}{\partial y^2} = 0 \quad (5)$$

- Analyzed using a Fourier series expansion:

$$u = \frac{1}{4\pi^2} \sum_j \sum_k \hat{u}_{jk} e^{i(\sigma_x)_j x} e^{i(\sigma_y)_k y} \quad (6)$$

- Yields:

$$A \left( \frac{\sigma_x}{\sigma_y} \right)^2 + B \left( \frac{\sigma_x}{\sigma_y} \right) + C = 0 \quad (7)$$

- Soon, we'll look at how this equation applies to PDEs.

# PDE Classification - Fletcher 1991

- An alternative approach that is more general (applies to  $N$  independent variables) and is provided by [Fletcher 1991] in [Versteeg and Malalasekera 1995]:

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial xy} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu + G = 0 \quad (8)$$

- This can be written in summation form, with  $A_{jk} = A_{kj}$  as:

$$\sum_{j=1}^N \sum_{k=1}^N A_{jk} \frac{\partial^2 u}{\partial x_j \partial x_k} + H = 0 \quad (9)$$

- According to Fletcher [Fletcher 1991], the eigenvalues of the A-matrix (resulting from the  $A_{jk}$ ) provide a result for PDE classification.

# PDE Classification - Fletcher 1991

- The classification rule [Fletcher 1991] is based on the Eigenvalues of  $[A]$ .
- Let's try this out for a PDE with  $N=2$  independent variables:

$$M = \begin{bmatrix} A & B \\ B & C \end{bmatrix} \quad (10)$$

- To find the eigenvalues of  $M$  we need to find  $\lambda$  that satisfy  $\det|M - \lambda I| = 0$ :

$$\det|M - \lambda I| = 0 \quad = \quad \det \begin{bmatrix} A - \lambda & B \\ B & C - \lambda \end{bmatrix}$$

# PDE Classification - Fletcher 1991

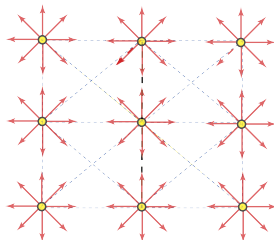
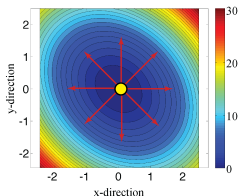
- The result from this approach [Fletcher 1991][Versteeg and Malalasekera 1995] is that the eigenvalues will classify the equation:
  - If any eigenvalue,  $\lambda = 0 \rightarrow$  **Parabolic Equation**
  - If all eigenvalues are the same sign **and**,  $\lambda \neq 0 \rightarrow$  **Elliptic Equation**
  - If all eigenvalues are the same sign but ONE **and**,  $\lambda \neq 0 \rightarrow$  **Parabolic Equation**

# PDE Classification: Elliptic PDE

- Start with Laplace's or Poisson's Equation

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial xy} + C \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 1 \cdot \frac{\partial^2 \phi}{\partial x^2} + 1 \cdot \frac{\partial^2 \phi}{\partial y^2} = 0$$

- So,  $A = C = 1$  and  $B = 0$ , so  $B^2 - 4AC = -4 < 0$



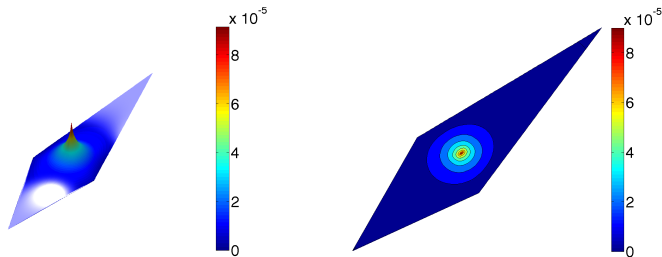
- Using [Fletcher 1991] approach:

$$M = \begin{bmatrix} A & B \\ B & C \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \lambda_1 = \lambda_2 = 1 \rightarrow \textit{Elliptic} \quad (11)$$

- Same sign, not equal to zero  $\rightarrow$  ELLIPTIC

# PDE Classification: Example

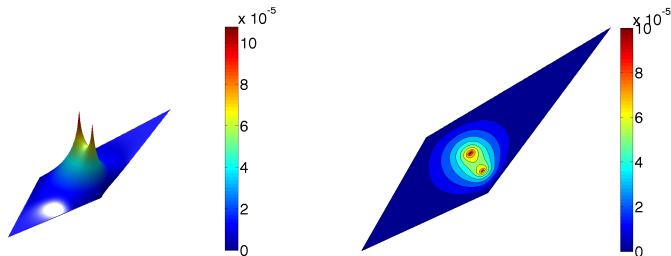
- Let's say you solve the membrane problem from last unit with a single point force applied transverse to the membrane:



- The effect of the point load is *distributed* throughout the membrane.
- The point has **influence** throughout the membrane.
- The node deflection(s) will **depend** on (all) the prescribed boundaries and internal nodes.

# PDE Classification: Example

- Let's add another point load:



- The point load has *influence* throughout the membrane.
- Deflection at any point is *dependent* on the forcing conditions at all points on the membrane and boundary.

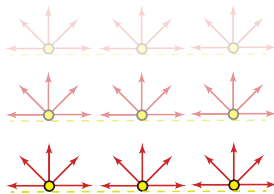
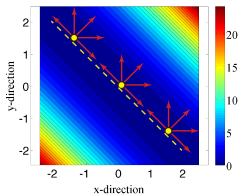


# PDE Classification: Parabolic PDE

- Next, let's consider the heat equation:

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial xy} + C \frac{\partial^2 u}{\partial y^2} = \frac{\partial T}{\partial t} - \frac{\partial^2 T}{\partial x^2} = f \quad (12)$$

- $A = 1, C = 0, B = 0$ . Therefore,  $B^2 - 4AC = 0$



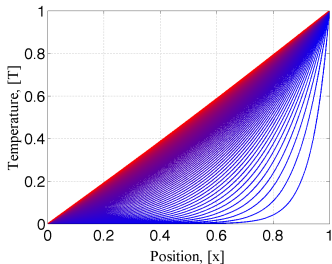
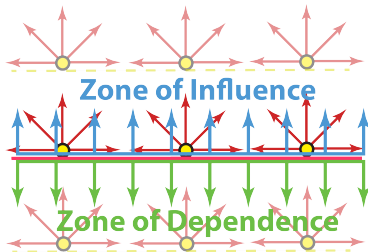
- Using [Fletcher 1991] approach:

$$M = \begin{bmatrix} A & B \\ B & C \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \lambda_1 = 1; \lambda_2 = 0 \rightarrow \textit{Parabolic} \quad (13)$$

- One eigenvalue is equal zero  $\rightarrow$  PARABOLIC

# PDE Classification: Example

- Consider the unsteady heat equation
- The temperature of the bar is **dependent** on the temperature in the time leading up to the current time.
- The temperature(s) now will **influence** only the temperature(s) in the future.
- I.e. If we turn up the heat now, it will only affect the temperature at times after the current time.

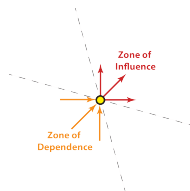
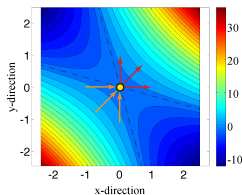


# PDE Classification: Hyperbolic PDE

- Next, let's consider the second order wave equation:

$$A \frac{\partial^2 u}{\partial t^2} + B \frac{\partial^2 u}{\partial t x} + C \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = f \quad (14)$$

- $A = 1, B = 0, C = -1$ . Therefore,  $B^2 - 4AC > 0$



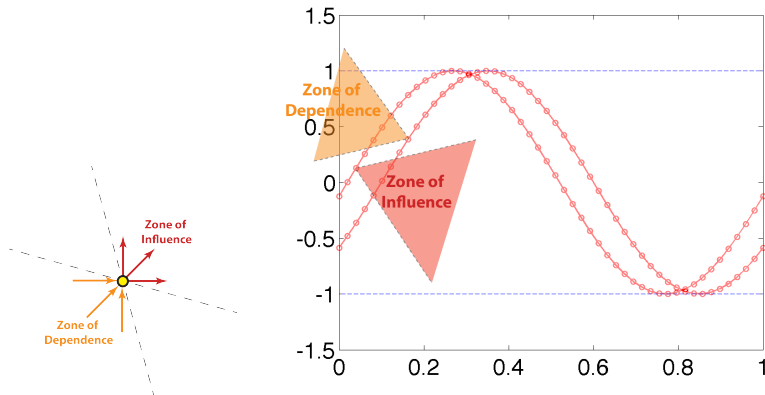
Using [Fletcher 1991] approach:

$$M = \begin{bmatrix} A & B \\ B & C \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \rightarrow \lambda_1 = 1; \lambda_2 = -1 \rightarrow \textit{Hyperbolic} \quad (15)$$

Eigenvalues have different signs, non-zero  $\rightarrow$  HYPERBOLIC

# PDE Classification: Example

- The wave equation propagates information to well defined zones of influence.



# PDE Classification: PDE Classification/Behavior Can Vary

- Consider the steady, isentropic, inviscid compressible flow Equation

$$(1 - M_\infty) \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = (1 - M_\infty) \cdot \frac{\partial^2 \phi}{\partial x^2} + 1 \cdot \frac{\partial^2 \phi}{\partial y^2} = 0$$

- So,  $A = (1 - M_\infty)$ ,  $C = 1$  and  $B = 0$ , so  $B^2 - 4(1 - M_\infty)C = ??$  will depend on the value of  $M_\infty$ .
- If  $M_\infty < 1$ , then ELLIPTIC
- If  $M_\infty > 1$ , then HYPERBOLIC
- Using [Fletcher 1991] approach:

$$M = \begin{bmatrix} A & B \\ B & C \end{bmatrix} = \begin{bmatrix} (1 - M_\infty) & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \lambda_1 = 1 - M_\infty, \lambda_2 = 1 \rightarrow ?? \quad (16)$$

- Again, the same holds true,  $M_\infty < 1 \rightarrow$  ELLIPTIC vs.  $M_\infty > 1 \rightarrow$  Hyperbolic.
- This example shows how the speed of sound is a key determinant into the behavior of compressible gas dynamics.
  - The zones of dependence and influence change with Mach number.
  - For subsonic flight, the aircraft will affect the surroundings in all

# What have we learned?

- A second order (partial differential) equation can be classified into 3 types:
  - **Elliptic:** Laplace and other similar equations. Elliptic equations have zones of influence and dependence that include the *whole* domain.
  - **Parabolic:** Unsteady heat, parabolized Navier-Stokes, boundary layers, and other similar equations. Parabolic equations have zones of influence that includes the *downstream* domain. The zone of dependence is the upstream domain.
  - **Hyperbolic:** Wave equation, supersonic Navier-Stokes, and other similar equations. Hyperbolic equations have zones of influence and dependence that converge to the point along characteristic lines (upstream & downstream).